

Two-Stage Adaptive and Compressed CSI Feedback for FDD Massive MIMO

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Abstract—Channel state information (CSI) feedback is a key technical challenge for frequency division duplex (FDD) massive multiple-input multiple-output (MIMO) systems. We propose a two-stage adaptive and compressed CSI feedback scheme to reduce the CSI feedback overhead and improve the feedback accuracy. Specifically, a binary angle-of-departure (AoD) vector and the associated angular-domain channel coefficients are first extracted from the massive MIMO channel. Then, by exploiting the dynamic burst-sparsity of the massive MIMO channel in the joint time-angular domain, a two-stage differential feedback scheme is proposed to compress and feed back the AoD vector, where the feedback overhead is adaptively adjusted according to the AoD changing rate. In addition, an adaptive-codebook-based feedback scheme is proposed for the angular-domain channel coefficients. Compared with the 5G New Radio (NR) Type-2 feedback scheme in 3GPP R15, the proposed scheme can significantly reduce the feedback overhead and achieve better block error rate (BLER) performance.

Index Terms—CSI feedback, compressive sensing, adaptive codebook.

I. INTRODUCTION

Massive multiple-input multiple-output (MIMO) is a core technology in future communication systems. To fully unleash the potential gain of massive MIMO, it is essential for the base station (BS) to obtain the downlink channel state information (CSI). In frequency division duplex (FDD) systems, the downlink CSI has to be obtained via CSI feedback from users.

Recently, several codebook based CSI feedback schemes have been proposed for FDD massive MIMO systems. In [1], a compressed sensing based codebook was proposed, which is only valid when channel statistics is stationary. An angle-of-departure (AoD) adaptive subspace codebook was proposed in [2], which can effectively reduce the feedback overhead by assuming constant AoD support within the angle coherence time. However, in realistic channels, the channel statistics/AoD support can change abruptly due to various practical issues, such as the blockage effect [3]. Several joint channel estimation and feedback methods were also proposed in [4]–[7]. For example, a multi-resolution hierarchical codebook and an adaptive estimation algorithm were proposed in [4] to achieve channel estimation at the BS. An angular domain differential strategy was proposed in [5] for the angle estimation in cross-polarized MIMO. The spatial sparsity of the self interference (SI) channel was exploited to suppress the interference between the users [6] and to avoid the pilot contamination [7]. However, these implicit feedback schemes in [4]–[7] do not provide a solution

for the user to directly feed back its estimated channel vector and is not compatible with the 5G NR protocol. In contrast, an enhanced CSI feedback scheme called NR Type-2 feedback has been proposed in 3GPP R15 to meet the improved CSI feedback accuracy required by the massive MIMO [8]. This CSI feedback scheme can handle the sudden change of the AoD support, but at the cost of much higher feedback overhead.

To overcome the drawbacks of the existing schemes, we propose a two-stage adaptive and compressed CSI (TSAC-CSI) feedback scheme, which can fully exploit the dynamic burst-sparsity of the massive MIMO channel and adaptively adjust the feedback overhead and codebook to significantly reduce the CSI feedback overhead with improved performance. Compared with the existing solutions, the proposed TSAC-CSI feedback scheme does not require restrictive assumptions, i.e., stationary channel statistics [1] or constant AoDs [2]. We conducted extensive simulations under realistic scenarios using LTE data path and clustered delay line (CDL) channel model in 3GPP R16 [9]. Both analysis and simulation results verify that our scheme could significantly reduce the feedback overhead while achieving better block error rate (BLER) performance compared to the NR Type-2 feedback scheme.

II. SYSTEM MODEL

A. Massive MIMO Downlink Transmission

Consider a massive MIMO system with one BS serving K users. The BS employs a uniform linear array (ULA) with $M \gg 1$ antennas and each user has N antennas, where the typical value for N is 2 or 4. In the massive MIMO downlink, each frame consists of three stages. In the downlink channel estimation stage, the BS transmits pilots for the users to estimate the downlink channels $\mathbf{H}_k \in \mathbb{C}^{N \times M}$, $k = 1, \dots, K$. In the CSI feedback stage, each user k feeds back CSI to the BS. In the downlink transmission stage, the BS performs MIMO precoding according to the feedback CSI to achieve a good downlink transmission performance. The exact CSI required by the BS depends on the specific precoding scheme. For example, the CSI could be the channel matrix $\mathbf{H}_k$ itself or the d largest eigenvalues and the associated eigenvectors of $\mathbf{H}_k^H \mathbf{H}_k$ in the SVD-based linear precoding scheme [10], where d is the number of data streams allocated to each user. In this paper, we will only focus on the CSI feedback stage by assuming that the channel matrix $\mathbf{H}_k$ is available at each user k . There are many existing works on massive MIMO channel estimation and the detailed estimation scheme is out of the scope of this paper.

B. Sparse CSI Feedback

Let $\mathbf{h}_{k,n}^H$ denote the n -th row of $\mathbf{H}_k$. Due to the limited number of scatterers, the massive MIMO channel has a sparse representation under an appropriate angular-domain sparse basis $\mathbf{D} \in \mathbb{C}^{M \times \bar{M}}$ as $\mathbf{h}_{k,n} = \mathbf{D} \mathbf{x}_{k,n}$ [11], where $\bar{M} \geq M$ is the size of the sparse basis $\mathbf{D}$, and $\mathbf{x}_{k,n} \in \mathbb{C}^{\bar{M} \times 1}$ is the sparse channel vector corresponding to $\mathbf{h}_{k,n}^H$. The sparse basis $\mathbf{D} = [\mathbf{d}_1, \dots, \mathbf{d}_{\bar{M}}]$ can be trained using the offline dictionary learning method in [11]. Note that the other form of CSI, such as the eigenvector of $\mathbf{H}_k^H \mathbf{H}_k$ also has a similar sparse representation under the same sparse basis $\mathbf{D}$, since a reasonable CSI for MIMO precoding should lie in the subspace spanned by $\mathbf{h}_{k,n}$, $n = 1, \dots, N$. Therefore, without loss of generality, we will focus on designing a feedback scheme for the channel vector $\mathbf{h}_{k,n}$ of user k , and the subscript k, n will be omitted for conciseness.

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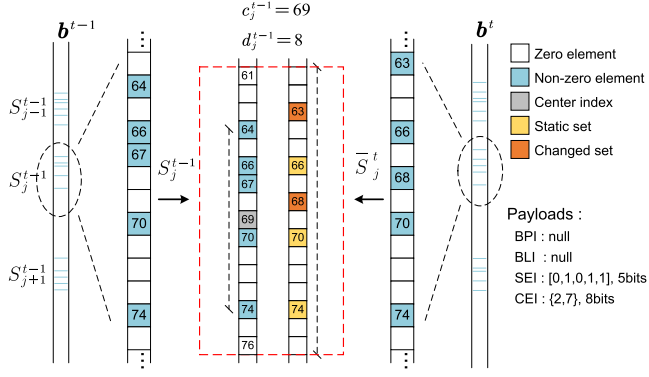


Fig. 1. Illustration of the burst-sparsity and strong time correlations for binary AoD vector.

Since $\mathbf{D}$ is fixed and can be determined in an offline manner as shown in [11]. The user only needs to feed back the sparse channel vector $\mathbf{x}$ to the BS. Specifically, define a binary AoD vector $\mathbf{b} = [b_1, \dots, b_M] \in \{0, 1\}^M$ such that $b_m = 0$ if x_m is zero (or close to zero in practice) and $b_m = 1$ otherwise. Based on $\mathbf{D}$ and the estimated channel matrix $\mathbf{H}$, the user can extract the AoD vector $\mathbf{b}$ and the corresponding non-zero sub-vector $\mathbf{z}$ in $\mathbf{x}$, using some existing algorithms such as the orthogonal matching pursuit (OMP) or the l_1 -norm minimization [12]. Then, the AoD vector $\mathbf{b}$ and the channel coefficients $\mathbf{z}$ are sent to the BS using some limited-feedback schemes. Finally, the BS approximately reconstructs the original channel vector $\mathbf{h}$ based on the limited feedback $\mathbf{b}$ and $\hat{\mathbf{z}}$, where the AoD feedback $\mathbf{b}$ can be perfect since it is binary and the channel coefficient feedback $\hat{\mathbf{z}}$ has quantization errors. In the following section, we shall exploit the dynamic burst-sparsity of the AoD vector $\mathbf{b}$ and the spatial correlation of the channel coefficients $\mathbf{z}$ to design an efficient TSAC-CSI feedback scheme.

III. PROPOSED TSAC-CSI FEEDBACK SCHEME

In this section, we develop a two-stage differential feedback scheme and an adaptive-codebook based quantization method to respectively compress the AoD support vector $\mathbf{b}$ and the channel coefficients $\mathbf{z}$. Throughout this section, we assume that the number of the non-zero elements of the AoD vector $\mathbf{b}$ is fixed as L . This is a reasonable assumption since in practice, the number of dominate paths usually changes very slowly and we can always choose a sufficiently large L (but still usually much less than M) such that most channel energy concentrates on L dominate paths for a long time.

A. Two-Stage Differential Feedback for AoD Vector

Many studies and experiments have shown that the massive MIMO channel exhibits burst-sparsity and strong time correlations due to the limited and slow-changing scatterers in the propagation environment [13]. Specifically, the non-zero elements of the AoD vector $\mathbf{b}$ concentrate on a few clusters and many elements in $\mathbf{b}$ remain the same in two consecutive frames, as shown in Fig. 1. In the following, such dynamic burst-sparsity is exploited to design a two-stage differential feedback scheme, which can significantly reduce the feedback overhead. The proposed scheme contains three steps, as elaborated below. Without loss of generality, we will focus on the feedback of the AoD vector in the t -th frame, denoted as $\mathbf{b}^t$.

1) *Step 1 (Segmentation of $\mathbf{b}^{t-1}$ at the BS and User Sides):* To exploit the dynamic burst-sparsity, in each frame t , the AoD vector

Algorithm 1: Iterative K-means for $\mathcal{P}_S$.

Input: AoD vector $\mathbf{b}^t$, ϵ , c_j^{t-1} , where c_j^{t-1} denotes the center index of segment $\mathcal{S}_j^{t-1}$ in $\mathbf{b}^{t-1}$.

Output: The index set of segments $\mathcal{S}_j^t$, $j = 1, \dots, N$.

Step 1 (Initialization): Initialize the segment number $N = 1$.

Step 2 (Iteration):

- **A:** Given the $\mathbf{b}^t$ and N , find a good segmentation solution based on K-means algorithm:
 - 1) Let $c_j^t = c_j^{t-1}$, $j = 1, \dots, N$, as the initial center indices of each segment $\mathcal{S}_j^t$. If $t = 1$, set $c_j^t = \lceil |\mathbf{b}^t|(2j-1)/2N \rceil$, $\forall j$.
 - 2) Repeat the following two sub-steps until each Ω_j^t has no change.
 - a) Assign each non-zero element in $\mathbf{b}^t$ to the segment Ω_j^t based on the minimum distance to c_j^t ;
 - b) Update $c_j^t = \lceil [\max(\Omega_j^t) + \min(\Omega_j^t)]/2 \rceil$ as the new center of each segment.
- **B:** Let $\mathcal{S}_j^t = \{i : \min(\Omega_j^t) \leq i \leq \max(\Omega_j^t)\}$, $\sigma = \min_{j=1, \dots, N} |\Omega_j^t|/|\mathcal{S}_j^t|$. If $\sigma + \epsilon \geq 1$, stop and output $\mathcal{S}_j^t$, $\forall j$; Else, set $N = N + 1$ and go to Step 2.A.2.

$\mathbf{b}^{t-1}$ is divided into J^{t-1} non-zero segments with $\mathcal{S}_j^{t-1}$ denoting the index set of the j -th segment, such that the total number of segments J^{t-1} is small and the portion of the non-zero elements in each segment is close to one. Specifically, letting Ω_j^{t-1} denote the index set of the non-zero elements of $\mathbf{b}^{t-1}$ in $\mathcal{S}_j^{t-1}$, and Ω^{t-1} denote the index set of all the non-zero elements of $\mathbf{b}^{t-1}$, the segmentation of $\mathbf{b}^{t-1}$ can be formulated as the following problem

$$\mathcal{P}_S : \min_{J^{t-1}, \{\mathcal{S}_j^{t-1}, \dots, \mathcal{S}_{J^{t-1}}^{t-1}\}} J^{t-1} \quad \text{s.t. } \Omega^{t-1} \subseteq \bigcup_{j=1}^{J^{t-1}} \mathcal{S}_j^{t-1}, \quad (1)$$

$$\frac{|\Omega_j^{t-1}|}{|\mathcal{S}_j^{t-1}|} \geq 1 - \epsilon, \forall j, \quad (2)$$

where $\epsilon > 0$ is a small number. Constraint (1) means that the J^{t-1} segments must contain all non-zero elements of $\mathbf{b}^{t-1}$. Constraint (2) means that the portion of the non-zero elements in each segment must be larger than $1 - \epsilon$. Problem $\mathcal{P}_S$ is a difficult set optimization problem and finding the optimal solution requires exhaustive search whose complexity is unacceptable in practice. Therefore, we propose an iterative K-means algorithm to find a good solution, as summarized in Algorithm 1. The basic idea is to use the K-means algorithm to cluster the non-zero elements into J^{t-1} segments for increasing J^{t-1} until Constraint (2) is satisfied. Note that both the BS and user run the same Algorithm 1 to obtain segmentation $\mathcal{S}_j^{t-1}$, $\forall j$, for $\mathbf{b}^{t-1}$, which serves as the shared information between the BS and user for reducing the feedback overhead of $\mathbf{b}^t$.

2) *Step 2 (Segmentation of $\mathbf{b}^t$ at the User Side):* Based on the shared information $\mathcal{S}_j^{t-1}$, $\forall j$, the user further divides $\mathbf{b}^t$ into $\bar{J}^t$ non-zero segments as follows. Formally, let $c_j^{t-1} = \lceil [\max(\Omega_j^{t-1}) + \min(\Omega_j^{t-1})]/2 \rceil$ and $d_j^{t-1} = 2^{\lceil \log_2 |\mathcal{S}_j^{t-1}| \rceil - 1}$ denote the center and radius of the j -th segment of $\mathbf{b}^{t-1}$. Note that the radius is set as an integer power of 2 to facilitate the quantization and feedback. Without extra computation, the j -th segment $\bar{\mathcal{S}}_j^t$ of $\mathbf{b}^t$ can be immediately determined as $\bar{\mathcal{S}}_j^t = \{i : c_j^{t-1} - d_j^{t-1} \leq i \leq c_j^{t-1} + d_j^{t-1} - 1\}$.

TABLE I
NOTATIONS OF THE INDICATORS

Name	1. Descriptions / 2. Overhead per-segment (bits)
BPI	1. <i>Burst position indicator</i> is used to feed back the center information of the last $\bar{J}^t - J^{t-1}$ segments. 2. $\lceil \log_2 \mathbf{b}^t \rceil$
BLI	1. <i>Burst length indicator</i> is used to quantize the length information for each segment. 2. $\lceil \log_2 \lceil \log_2 \mathbf{b}^t \rceil \rceil$
SEI	1. <i>Static set indicator</i> is a binary vector, through which the static set can be accurately fed back. 2. $ \Omega_j^{t-1} $
CEI	1. <i>Changed set indicator</i> is used to directly quantize the changed set. 2. $ \Omega_j^t - \Omega_j^{t-1} \cap \Omega_j^t \times \lceil \log_2 \mathcal{S}_j^{t-1} \rceil$

Then given $\bar{\mathcal{S}}_j^t, j = 1, \dots, J^{t-1}$, we first remove all the non-zero elements in $\cup_{j=1}^{J^{t-1}} \bar{\mathcal{S}}_j^t$ to obtain a residue AoD vector $\bar{\mathbf{b}}^t$, and then we apply Algorithm 1 to divide the residue AoD vector $\bar{\mathbf{b}}^t$ into $\bar{J}^t - J^{t-1}$ segments as $\bar{\mathcal{S}}_j^t, j = J^{t-1} + 1, \dots, \bar{J}^t$, where $\bar{J}^t$ is also determined by Algorithm 1. The final segmentation of $\mathbf{b}^t$ is given by $\bar{\mathcal{S}}_j^t, j = 1, \dots, \bar{J}^t$. Note that the first J^{t-1} segments of $\mathbf{b}^t$ are obtained based on $\mathcal{S}_j^{t-1}, \forall j$ instead of directly using Algorithm 1 because we want to exploit the shared information to reduce the feedback overhead. Furthermore, we use different notations $\mathcal{S}_j^t$ and $\bar{\mathcal{S}}_j^t$ to denote the shared segmentation information and the segmentation at user side, respectively, since they are calculated using different methods for different purposes.

3) *Step 3 (Two-Stage Per-Segment AoD Feedback)*: In this step, four indicators will be introduced to execute a two-stage AoD feedback for accurately quantizing the indices of non-zero elements in each segment $\bar{\mathcal{S}}_j^t$. The detail descriptions of these indicators are tabulated in Table I and the two-stage feedback scheme is summarized below.

In the first stage, BPI and BLI are used to first feed back the center and radius information c_j^{t-1}, d_j^{t-1} of the last $\bar{J}^t - J^{t-1}$ segments, whose quantization overhead are $\lceil \log_2 |\mathbf{b}^t| \rceil$ bits and $\lceil \log_2 \lceil \log_2 |\mathbf{b}^t| \rceil \rceil$ bits per-segment, respectively. Note that the overhead of BLI is $\lceil \log_2 \lceil \log_2 |\mathbf{b}^t| \rceil \rceil$ bits because d_j^{t-1} is chosen from $\lceil \log_2 |\mathbf{b}^t| \rceil$ different values, since we have defined the radius as an integer power of 2. After the first stage feedback, the BS will retrieve the segmentation information $\bar{\mathcal{S}}_j^t, j = 1, \dots, \bar{J}^t$.

In the second stage, the indices of non-zeros elements in each segment $\bar{\mathcal{S}}_j^t$ will be quantized and fed back. Specifically, the user first divides the indices of the non-zero elements per-segment $\bar{\mathcal{S}}_j^t$ into two sets: the static set $\Omega_j^{t-1} \cap \Omega_j^t$ and changed set $\Omega_j^t - \Omega_j^{t-1} \cap \Omega_j^t$. Then, a length- $|\Omega_j^{t-1}|$ binary vector, called SEI, is used to feed back the static set, such that elements one indicate the static elements in Ω_j^{t-1} . Finally, the order indices of the changed set in $\bar{\mathcal{S}}_j^t$ are directly quantized via CEI, which costs $\lceil \log_2 |\bar{\mathcal{S}}_j^t| \rceil$ bits for each element.

Based on the proposed two-stage feedback scheme, the AoD vector $\mathbf{b}^t$ can be perfectly reconstructed at the BS with a small feedback overhead, thanks to the shared information $\mathcal{S}_j^{t-1}, \Omega_j^{t-1}, \forall j$ generated by exploiting the strong correlations between the AoD vectors in two consecutive frames. Note that when the AoD vector changes at a faster rate, the feedback overhead will be larger since both $\bar{J}^t - J^{t-1}$ and $|\Omega_j^t - \Omega_j^{t-1} \cap \Omega_j^t|$ will become larger. Therefore, the feedback

overhead is adaptively adjusted according to the AoD changing rate. To facilitate analysis, an example that illustrates the quantization process of a single segment is shown in Fig. 1 and the payloads of the four indicators are also listed. For the general case, the feedback overhead is given by

$$F_b = \Delta + \sum_{j=1}^{\bar{J}^t} [\lceil \Omega_j^{t-1} \rceil + |\Omega_j^t - \Omega_j^{t-1} \cap \Omega_j^t| \times \lceil \log_2 |\mathcal{S}_j^{t-1}| \rceil], \quad (3)$$

where $\Delta = (\bar{J}^t - J^{t-1}) \times (\lceil \log_2 |\mathbf{b}^t| \rceil + \lceil \log_2 \lceil \log_2 |\mathbf{b}^t| \rceil \rceil)$ is the feedback overhead of first stage.

B. Adaptive-Codebook Based Channel Coefficients Feedback

The proposed channel coefficients feedback scheme includes two key components: the codebook-based multi-atom vector quantization and the mixed offline-online codebook learning, as elaborated below.

1) *Codebook-Based Multi-Atom Vector Quantization*: Given the codebook $\mathcal{C} = [\mathbf{c}_1, \dots, \mathbf{c}_{2^B}] \in \mathbb{C}^{L \times 2^B}$ with 2^B atoms (codevectors), we propose a multi-atom vector quantization scheme for $\mathbf{z}$, which contains two steps. In the first step, the user finds a multi-atom representation of $\mathbf{z}$ as $\mathbf{z} = \sum_{i \in \mathcal{I}^*} p_i \mathbf{c}_i + \mathbf{e}$, where $\mathbf{e}$ is the representation error, and $\mathcal{I}^*$ is obtained by solving the following problem:

$$\mathcal{P}_Z : \min_{\mathcal{I} \subseteq \{1, \dots, 2^B\}} \left\| \mathbf{z} - \sum_{i \in \mathcal{I}} p_i \mathbf{c}_i \right\|^2, \text{ s.t. } |\mathcal{I}| = l, \quad (4)$$

where l is the number of atoms used to represent $\mathbf{z}$. Problem $\mathcal{P}_Z$ is again a set optimization problem which is difficult to solve directly. To overcome this challenge, we transform $\mathcal{P}_Z$ into the following equivalent problem:

$$\mathcal{P}'_Z : \min_{\tilde{\mathbf{p}}} \|\mathbf{z} - \mathcal{C}\tilde{\mathbf{p}}\|^2, \text{ s.t. } \|\tilde{\mathbf{p}}\|_0 = l, \quad (5)$$

where $\tilde{\mathbf{p}} \in \mathbb{C}^{2^B \times 1}$ is the spare representation of the coefficients $\mathbf{z}$. $\mathcal{P}'_Z$ is a standard formulation in compressive sensing (CS) and many existing CS recovery algorithms, such as OMP [14], can be used to find a good solution. The details are omitted for conciseness.

In the second step, the user sends the set $\mathcal{I}^*$ and the quantized coefficients $\mathbf{p}$ to the BS. Specifically, $\lceil \log_2 (C_{2^B}^l) \rceil$ bits are required to represent the set $\mathcal{I}^*$ and m, n bits are used to quantify the amplitude and phase for each element of $\mathbf{p}$, respectively. Therefore, the total number of feedback bits for $\mathbf{z}$ is given by

$$F_z = \lceil \log_2 (C_{2^B}^l) \rceil + l \times (m + n), \quad (6)$$

where typical codebook size is determined by L and $B = 7$ or 8 are commonly used in practice. Note that F_z can be further compressed as $F_{\bar{z}}$ if the BS only needs the channel direction information, i.e., $\mathbf{z}/\|\mathbf{z}\|$, where

$$F_{\bar{z}} = \lceil \log_2 (C_{2^B}^l) \rceil + \lceil \log_2 l \rceil + (l - 1) \times (m + n). \quad (7)$$

In this paper, we configure two feedback modes: low resolution feedback mode (LRFM) with $\{l, m, n\} = \{2, 3, 3\}$ and high resolution feedback mode (HRFM) with $\{l, m, n\} = \{3, 4, 4\}$. Simulations in section IV show that the proposed multi-atom vector with $l > 1$ can achieve a more flexible tradeoff between the quantization error and feedback overhead.

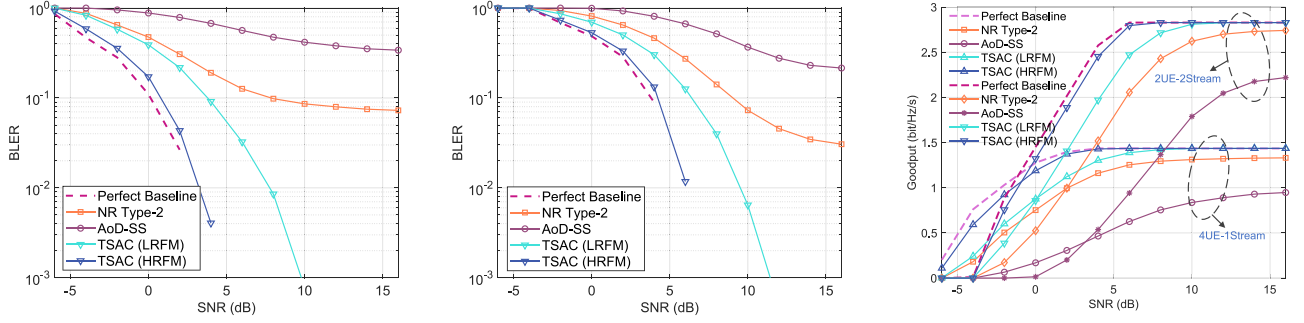


Fig. 2. Simulation results. (a) BLER performance of 4UE-1stream scenario; (b) BLER performance of 2UE-2stream scenario; (c) Goodput performance of the both scenarios.

2) *Mixed Offline-Online Codebook Learning*: We design a mixed offline-online codebook learning method to increase the representation accuracy of (5). Both offline training and online updating can be formulated as a dictionary learning problem:

$$\mathcal{P}_C : \min_{\mathbf{C}, \tilde{\mathbf{P}}} \left\| \mathbf{Z} - \mathbf{C}\tilde{\mathbf{P}} \right\|_F^2, \text{ s.t. } \|\tilde{\mathbf{p}}_i\|_0 \leq l, \forall i, \quad (8)$$

where subscript F represents Frobenius norm, $\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_K]$ are K experienced channel coefficient vectors (training data), $\tilde{\mathbf{P}} = [\tilde{\mathbf{p}}_1, \tilde{\mathbf{p}}_2, \dots, \tilde{\mathbf{p}}_K]$ and $\tilde{\mathbf{p}}_k$ is the sparse representation of $\mathbf{z}_k$. To reduce the complexity, we transform $\mathcal{P}_C$ into the following clustering problem $\mathcal{P}'_C$:

$$\begin{aligned} \mathcal{P}'_C : \min_{\mathbf{c}, \{\mathcal{Z}^i, i=1, \dots, 2^B\}} & \sum_{i=1}^{2^B} \sum_{\mathbf{z}_j \in \mathcal{Z}^i} \|\mathbf{c}_i - \mathbf{z}_j\|^2 \\ \text{s.t. } & \bigcup_{i=1}^{2^B} \mathcal{Z}^i = \mathcal{Z}, \\ & \mathcal{Z}^p \cap \mathcal{Z}^q = \emptyset, \quad \forall p, q \in \{1, \dots, 2^B\}, \end{aligned} \quad (9)$$

where $\mathcal{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_K\}$ is the training set, which is divided into 2^B clusters $\mathcal{Z}^i$'s around the 2^B codevectors $\mathbf{c}_i$'s. Problem $\mathcal{P}'_C$ can be easily solved by existing clustering algorithms, such as Linde-Buzo-Gray (LBG) [15]. Before the online operation of the system, the BS can train an appropriate initial codebook by solving $\mathcal{P}'_C$ with some offline training data $\mathcal{Z}$ and then send the initial codebook to the users. Moreover, during the online operation stage, after observing the channel coefficient vector $\hat{\mathbf{z}}^t$ at each frame t , the codebook can be synchronously updated (using e.g., the online version of the LBG) at both BS and users to track the varying channel statistics as

$$\mathbf{c}_i^{t+1} = (\tau_i^t \mathbf{c}_i^t + (1 - \tau_i^t) \hat{\mathbf{z}}^t) / \|\tau_i^t \mathbf{c}_i^t + (1 - \tau_i^t) \hat{\mathbf{z}}^t\|, \quad (10)$$

where $\mathbf{c}_i^t$ denotes the nearest codevector to $\hat{\mathbf{z}}^t$ at frame t , $\mathbf{c}_i^{t+1}$ denotes the updated codevector and τ_i^t is the step size. It is noteworthy that the updated codebook at the BS and user sides remain the same because they use the same data $\hat{\mathbf{z}}^t$ to update the codebook at each frame t . Meanwhile, the online computational overhead is small since only one codevector around $\hat{\mathbf{z}}^t$ will be updated at each frame t .

IV. SIMULATION RESULTS

In this section, We conduct extensive link-level simulations using CDL-B channel model to verify the superior performance of the proposed scheme over the NR Type-2 feedback scheme [16]. Note that

since the more realistic CDL-B channel model does not satisfy the assumption of constant channel statistics or AoDs over a larger number of frames, the CSI feedback scheme in [1], [2] cannot be directly applied. Therefore, we consider a genie-aided version of the AoD-adaptive subspace codebook based CSI feedback (AoD-SS) scheme in [2], where the BS is assumed to have the perfect information about the channel support. This baseline scheme serves as a performance upper bound for the AoD-SS scheme in [2]. We consider dual-polarized antenna, and the number of transmit antennas M , the number of receive antennas N per user, and the number of users K are $(M, N, K) = (32, 4, 4)$. The delay spread and UE speed are set as 1 us and 3 km/h respectively. The sparse basis $\mathbf{D}$ is chosen to be a discrete Fourier transform (DFT) matrix. In addition, we consider a multi-layer transmission where the modulation and coding scheme (MCS) and the number of physical resource block (PRB) are 8, 24 respectively [17].

Fig. 2(a) and Fig. 2(b) show the average BLER performance per-user for each scheme. Based on (3), (6) and Reference [16], the average feedback overhead per-stream is 32.1 bits ($F_b = 11.1$ bits, $F_z = 24$ bits) for LRFM, 51.7 bits for HRFM ($F_b = 11.7$ bits, $F_z = 40$ bits), 61 bits for NR Type-2 scheme and 12 bits for the genie-aided AoD-SS. Note that for the genie-aided AoD-SS, 12 feedback bits already correspond to a codebook of size 2^{12} and further increasing the number of feedback bits would lead to an unacceptable complexity in practice. Therefore, the single-atom vector quantization in [2] is not suitable for high accuracy feedback requirement in massive MIMO. Compared to the NR scheme, the proposed LRFM can achieve the same target average BLER of 0.1 with a much lower SNR and only about half of the feedback overhead. Compared to the genie-aided AoD-SS, proposed LRFM can achieve much better performance with only 12 additional feedback bits for channel coefficients. In addition, proposed HRFM can approach the ideal baseline of perfect CSIT, still with less feedback overhead than the NR scheme.

Fig. 2(c) compare the average goodput performance per-user of different schemes, where goodput $\triangleq R \times (1 - \text{mean}(\text{BLER}))$ and R denotes the transmission rate. We can observe that the goodput gap between proposed HRFM and the ideal CSIT is much smaller than that with NR Type-2 scheme. In summary, the above simulation results verify the better transmission performance of proposed scheme with less feedback overhead. The large performance gain is due to the following reasons. First, the proposed method exploits the burst-sparsity and time correlations of the AoD support to reduce the feedback overhead. Second, with the mixed offline-online codebook learning, the proposed multi-atom vector quantization method outperforms the scalar quantization of NR scheme and conventional vector quantization of genie-aided AoD-SS.

V. CONCLUSION

In this paper, we proposed a two-stage adaptive and compressed CSI feedback scheme for FDD massive MIMO systems. By leveraging the dynamic burst-sparsity of the massive MIMO channel in the joint time-angular domain, a binary AoD vector and the associated angular-domain channel coefficients were first extracted, compressed and then fed back to the BS, respectively. Specifically, we proposed a two-stage differential feedback scheme for the compression of the AoD vector, where the feedback overhead can be adaptively adjusted according to the AoD changing rate. Moreover, an adaptive-codebook-based feedback scheme was proposed for the feedback of the angular-domain channel coefficients. Numerical simulation results verified that our scheme significantly outperforms the 5G NR Type-2 feedback scheme in both feedback overhead and quantization performance.

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